

Parametric form of a function or relation — Problem Set

Warm-up

1. Find the Cartesian equation for the curve with parametric equations $x = 2t$ and $y = 3t - 1$.
2. Eliminate the parameter t to find the Cartesian equation of $x = t + 1$, $y = t^2$.
3. Convert the parametric equations $x = 3t$ and $y = \frac{2}{t}$ into Cartesian form.
4. Given the parametric equations $x = \cos \theta$ and $y = \sin \theta$, find the corresponding Cartesian equation.
5. A curve is defined by $x = t^2$ and $y = 2t$. Find the coordinates of the point on the curve where $t = 3$.
6. Express y in terms of x for the parametric equations $x = 4 - t$ and $y = 2t + 5$.
7. Find the Cartesian equation of the curve represented by $x = 2 \sin \theta$ and $y = 2 \cos \theta$.
8. Determine the value of the parameter t at the point $(2, 4)$ for the curve $x = t$, $y = t^2$.
9. Eliminate the parameter t from the equations $x = 2^t$ and $y = 2^{2t}$.
10. Find the Cartesian equation for the line defined parametrically by $x = t - 3$ and $y = 3t$.

Skill-building

1. Find the Cartesian equation of the curve defined by $x = 3 \cos \theta - 1$ and $y = 3 \sin \theta + 2$, and state its geometric shape.
2. A curve is given by $x = t^2 - 2$ and $y = 2t + 1$. Find the coordinates of its points of intersection with the line $y = x + 1$.
3. Eliminate the parameter θ from the equations $x = 2 + \cos \theta$ and $y = 1 + \cos 2\theta$ to find the Cartesian equation.
4. Find the Cartesian equation for the curve $x = t^4$, $y = t^2$, and state any restrictions on x and y given that $t \geq 0$.
5. Eliminate the parameter t from the quadratic parametric equations $x = 2t^2 - 1$ and $y = t - 3$.
6. A curve is defined by $x = 4 \sin^2 \theta$ and $y = 4 \cos \theta$. Find the Cartesian equation and state the domain of the resulting function.
7. Find the Cartesian equation of the circle given by $x = -1 + 5 \cos \theta$ and $y = 4 + 5 \sin \theta$, explicitly stating its centre and radius.
8. Find the values of t where the parametrically defined curve $x = 2t$, $y = t^2 - 3t$ intersects the x -axis.

9. Express the Cartesian relation $y = 3x^2 - 2x + 1$ in parametric form using the parameter substitution $x = 2 - t$.

Easier Exam Questions

1. [Penrith (2023) Q1]

Find the Cartesian equation for the curve with the parametric equations:

$$x = 4p - 6$$

$$y = 3p$$

- (A) $y = \frac{1}{12}x + \frac{1}{2}$
(B) $y = \frac{3}{4}x + \frac{9}{2}$
(C) $y = \frac{1}{12}x - \frac{1}{2}$
(D) $y = \frac{3}{4}x + 6$

2. [Glenwood (2023) Q8f]

Find the Cartesian equation of the curve represented by the parametric equations $x = 5 - 2t$,
 $y = t^2 - t$ 2

3. [Caringbah (2019) Q1]

1 What is the Cartesian equation of the line with parametric equations

$$x = t + 2 \quad \text{and} \quad y = 2t + 1?$$

- (A) $2x + y + 3 = 0$
(B) $2x + y - 3 = 0$
(C) $2x - y + 3 = 0$
(D) $2x - y - 3 = 0$

4. [Baulkham Hills (2024) Q2]

A parabola has the parametric equations $x = t - 2$ and $y = 2t^2 - 1$. What is the Cartesian equation of the parabola?

- A. $y = 2x^2 + 4x - 1$
B. $y = 2x^2 + 4x + 1$
C. $y = 2x^2 + 8x - 7$
D. $y = 2x^2 + 8x + 7$

5. [Baulkham Hills (2019) Q7]

7 The parametric equations of a function are $x = 2t^2$, $y = 4 - t$. The Cartesian equation is;

- (A) $x = 4(2 - y)^2$
(B) $x = 4(y + 2)^2$
(C) $x = 2(y + 4)^2$
(D) $x = 2(4 - y)^2$

6. [Baulkham Hills (2020) Q11a]

The parametric equations of a curve are $x = 4 - t$, $y = (1 - t)^2$. Find the Cartesian equation of the curve, simplifying your answer. 2

7. [North Sydney Girls (2019) Q6e]

The point $P(10, -3)$ lies on the curve defined by the parametric equations: $x = 3t^2 - 2, y = 2t + 1$. What are the coordinates of the point Q whose parameter value is three less than the parameter value for P ? 2

8. [trimmed James Ruse (2020) Q4]

A possible set of parametric equations for the curve $4x^2 + y^2 = 24x - 2y - 33$ is

- (A) $x = 3 - \sin(t) \ y = 1 + 2 \cos(t)$
- (B) $x = 3 + \sin(t) \ y = -1 + 2 \cos(t)$
- (C) $x = 2 + \sin(t) \ y = 1 + 2 \cos(t)$
- (D) $x = -3 + \sin(t) \ y = 1 + 2 \cos(t)$

Harder Exam Questions

1. [trimmed International Grammar (2020) Q12(d)(i)]

A curve in the xy-plane is defined parametrically by $x = 4t^2$ and $y = 8t$, where t is the parameter.

Draw the curve for $-3 \leq t \leq 3$, showing the beginning and ending coordinates and any intercepts. 2

[trimmed International Grammar (2020) Q12(d)(ii)]

For what value or values of t will the parabola intersect the line $y = 6 - 2x$? 3

2. [Presbyterian (2023) Q11d]

A curve in the xy-plane is defined parametrically by $x = 8t$ and $y = 4t^2$, where t is the parameter.

(i) Draw the curve for $-3 \leq t \leq 3$, showing the co-ordinates of the endpoints and any intercepts. 2

(ii) For what value(s) of t will the curve intersect the line $y = 10 - \frac{3}{4}x$? 3

3. [Penrith (2023) Q13b]

Elena is standing in a field playing with toy remote control car. She is controlling the car so that it is driving in a circle around herself. The centre of the football field is taken to be the origin $(0,0)$. The position of a car in relation to the original is given by the equation $x^2 + 4x + y^2 - 5y = 20$ where x is the horizontal displacement and y is the vertical displacement in metres from the origin.

(i) Show that Elena is standing at the coordinate $(-2, 2.5)$ on the football field. 1

(ii) Show that the parametric equations that represent the horizontal and vertical displacements in terms of the angle, θ degrees, from the Elena are: 3

$$x = \frac{11 \cos \theta - 4}{2}$$

$$y = \frac{11 \sin \theta + 5}{2}$$

(iii) Find the position on the football field of the toy car when it is on an angle of 135° from Elena. Give your answer as a coordinate in simplified exact form. 2

4. [Penrith (2023) Q10]

The Cartesian equation $y = 2x^2 - 5x + 3$, is being converted to parametric equations with

$x = 5 - 3m$. Find the parametric equation for y in terms of m .

- (A) $y = 18m^2 - 15m + 28$
- (B) $y = 18m^2 - 45m + 28$
- (C) $y = 18m^2 + 15m + 28$
- (D) $y = 18m^2 + 45m + 28$

5. [Knox (2023) Q13f(i)]

(f) A curve has parametric equations $x = 1 + \cos \theta$ and $y = \cos 2\theta$ (i) Find the Cartesian equation of this curve. 1

[Knox (2023) Q13f(iii)]

(iii) What is the domain of this function. 1

6. [Girraween (2023) Q6c]

A curve is represented by the parametric equations:

$$x = \cos \theta + \sin \theta, y = \cos \theta - \sin \theta.$$

Find the Cartesian equation of the curve. 2

7. [Fort St (2020) Q8c(i)]

A curve has parametric equations $x = 4 \sin t + 3, y = 4 \cos t - 1$. 2

Eliminate the parameter t and show that the curve is a circle and give its Cartesian equation.

[Fort St (2020) Q8c(ii)]

Find the exact co-ordinates of the point on the curve where $t = \frac{\pi}{4}$. 2

8. [Fort St (2019) Q1d]

(i) Find the Cartesian equation of the curve defined by the parametric equations: $x = 2 \sin \theta$, $y = 3 + \cos 2\theta$ 2

(ii) State the type of graph the Cartesian equation represents. 1

9. [Baulkham Hills (2025) Q11d]

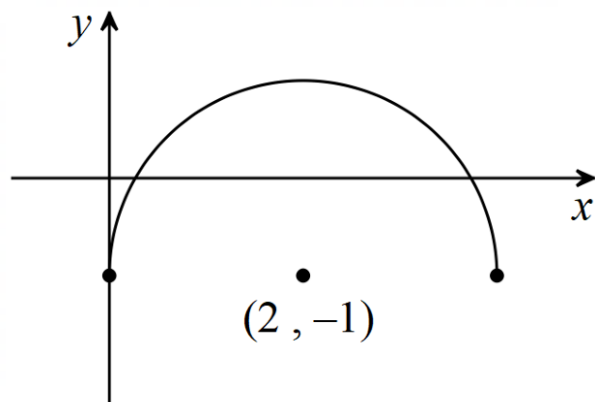
A curve is defined by the following parametric equations:

$$x = -3 + 2 \cos^2 \theta \quad \text{and} \quad y = -5 + \cos 2\theta$$

Find the Cartesian equation of the curve, noting any restrictions. 3

10. [Baulkham Hills (2022) Q7]

The diagram below shows an upper semicircular arc of radius 2, centred at $(2, -1)$. Which of



the following pair of parametric equations represents this arc?

- (A) $x = 2 \cos \theta + 2, y = 2 \sin \theta - 1$ from $\theta = 0$ to $\theta = \pi$.
- (B) $x = 2 \sin \theta + 2, y = 2 \cos \theta - 1$ from $\theta = 0$ to $\theta = \pi$.
- (C) $x = 2 \cos \theta + 2, y = 2 \sin \theta + 1$ from $\theta = 0$ to $\theta = \pi$.
- (D) $x = 2 \sin \theta + 2, y = 2 \cos \theta + 1$ from $\theta = 0$ to $\theta = \pi$.